

20250217EB_MN (qPCR)

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- 2025年2月17日 星期一
- split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs
- 2025年2月19日 星期三
- T2 -> T3
- 2025年2月21日 星期五
- T3 -> T4 (~14 mL for over-weekend)
- 2025年2月24日 星期一
- T4 -> T5
- 2025年2月26日 星期三
- T5 -> T6
- 2025年2月28日 星期五
- T6 -> T6 (~14mL for over-weekend)
- 2025年3月3日 星期一
- T6 -> T7
- Dissociation
1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs

2. Remove the PLO and wash the plate with PBS once

3. Coat the plate with laminine for 1hr.

4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).

5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.

6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)

7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)

8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.

9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

*I pipetted the cells very gentle this time

Table3					^
	A	B	C	D	
1	conc. (M/mL)	survival	total (5mL)	M/well (into 9*6-well)	
2	2.17	100%	10.85	1.2	

Well1				^
	1	2	3	
A	BP 1h +24hR	hypo1h +24hR	hypo24h+24hR	
B	BP 1h +24hR	hypo1h +24hR	hypo24h+24hR	

Well2				^
	1	2	3	
A	BP 1h	hypo 1h	hypo 24h	
B				

2025年3月5日 星期三

T7-> T7

2025年3月6日 星期四

T7 -> BP T8

2025年3月8日 星期六

-700uL + 800uL BP T8

2025年3月10日 星期一

700uL BP T8 -> 800 uL BP T8

2025年3月12日 星期三

700uL BP T8 -> 800 uL BP T8

*treat well1 A3&B3 with hypo (11:30am) (3:30pm)

2025年3月13日 星期四

well1: A1,2 & B1,2 -> BP/hypo 1h (10:30am) (2:30pm) -> rescue (11:30am) (3:30 pm)

well2: hypo at 11:30 am (3:30pm)

2025年3月14日 星期五

well 2: A1,2 BP/hypo (10:30am) (2:30pm)

all: freeze at (11:30) 3:30pm

2025年3月21日 星期五

RNA extraction with TRIzol (protocol)

Lyse samples and separate phases

1. Lyse and homogenize samples in TRIzol™ Reagent
2. Cell grown in monolayer:
 - a. Remove growth media.
 - b. Add 1 mL of TRIzol™ Reagent per 1×10^5 — 1×10^7 cells directly to the 3.5-cm culture dish to lyse the cells.
(The trizol froze when I added it to the plate which I stored in -80)
 - c. Pipet the lysate up and down several times to homogenize (I put the plate on the bench in RT, pipetted it, and let it thaw on the plate)
3. (Optional, I didn't do) If samples have a high fat content, centrifuge the lysate for 5 minutes at $12,000 \times g$ at 4 – 10°C , then transfer the clear supernatant to a new tube. 3. Incubate for 5 minutes to allow complete dissociation of the nucleoproteins complex.
4. Add 0.2 mL of chloroform per 1 mL of TRIzol™ Reagent used for lysis, securely cap the tube, then thoroughly mix by shaking.
5. Incubate for 2–3 minutes.
6. Centrifuge the sample for 15 minutes at $12,000 \times g$ at 4°C . The mixture separates into a lower phenol-chloroform, an interphase, and a colorless upper aqueous phase.
7. Transfer the aqueous phase containing the RNA to a new tube by angling the tube at 45° and pipetting the solution out.
(can only get 500uL/tube)
8. Precipitate the RNA
 - a. (Optional, I didn't do) If the starting sample is small ($<10^6$ cells or <10 mg of tissue), add 5–10 μg of RNase-free glycogen as a carrier to the aqueous phase. Note: The glycogen is co-precipitated with the RNA, but does not interfere with subsequent applications.
 - b. Add 0.5 mL of isopropanol (1 volume) to the aqueous phase, per 1 mL of TRIzol™ Reagent used for lysis.
 - c. Incubate for 10 minutes at 4°C . (incubated in -20 o/n instead to increase the yield)

==== the rest is on 3/22/2025=====

- a. Centrifuge for 10 minutes (did 30 min instead) at $12,000 \times g$ at 4°C . Total RNA precipitate forms a white gel-like pellet at the bottom of the tube.
 - b. Discard the supernatant with a micropipettor.
1. Wash the RNA
 - a. Resuspend the pellet in 1 mL of 75% ethanol per 1 mL of TRIzol™ Reagent used for lysis. Note: The RNA can be stored in 75% ethanol for at least 1 year at -20°C , or at least 1 week at 4°C .
 - b. Vortex the sample briefly, then centrifuge for 5 minutes at $7500 \times g$ at 4°C .
 - c. Discard the supernatant with a micropipettor.

- d. Vacuum or air dry the RNA pellet for 5–10 minutes. (I did air dry with tubes upside down. a bit more than 30 minutes in total) IMPORTANT! Do not dry the pellet by vacuum centrifuge. Do not let the RNA pellet dry, to ensure total solubilization of the RNA. Partially dissolved RNA samples have an A230/280 ratio <1.6.
2. Solubilize the RNA
- a. Resuspend the pellet in 20–50 µL (I used 18uL) of RNase-free water, 0.1 mM EDTA, or 0.5% SDS solution by pipetting up and down. IMPORTANT! Do not dissolve the RNA in 0.5% SDS if the RNA is to be used in subsequent enzymatic reactions.
- b. Incubate in a water bath or heat block set at 55–60°C for 10–15 minutes. Proceed to RT and store the rest of RNA at –80°C.

2025年3月22日 星期六

RT with Thermo Scientific Maxima™ H Minus cDNA Synthesis Master Mix with dsDNase

Table1			
	A	B	C
1	10X dsDNase Buffer	1 µL	
2	dsDNase	1 µL	
3	Template RNA •Total RNA or •poly(A) mRNA or •Specific RNA	1 pg to 5 µg 0.1 pg to 500 ng 0.01 pg to 500 ng	
4	Water, nuclease-free	to 12 µL	
5	Total volume	12 µL	

1. Prepare mix of
- a. 1ul DNase/sample +1
- b. 1ul of buffer 10x/sample +1
2. Add 2 ul/ tube. (add 2.2uL this time)
3. Mix gently and centrifuge.
4. Incubate at 37°C in a preheated thermomixer or water bath for 2 minutes.
5. Chill on ice, briefly centrifuge and place on ice.
6. Add the following components to the tube on ice:

Table2				^
	A	B	C	
1	Maxima cDNA H Minus Synthesis Master Mix (5X)	4 µL		
2	Water, nuclease-free	4 µL		

1. Mix gently and centrifuge.
2. Set up PCR program:

Table6			^
	A	B	
1	temp	time	
2	25	10	
3	50	15	
4	85	5	

- a. Incubate at 25 °C for 10 minutes.
- b. Incubate at 50 °C for 15 minutes. Note. If using >1 µg RNA template, increase the reaction time to 30 minutes.
For RNA templates that are GC-rich or have a large amount of secondary structure, the reaction temperature can be increased to 65 °C.
- c. Terminate the reaction by heating at 85 °C for 5 minutes.

***The reaction product of the first-strand cDNA synthesis can be used directly in qPCR. Store at –20°C for up to one week or –70°C for long-term storage. Avoid freeze-thaw cycles of the cDNA.

Table4										
	A	B	C	D	E	F	G	H	I	J
1	#	batch	Sample	RNA conc (ng/ul)	A260/280	A260/230	RNA amount for 10uL (ng)	minimum amount of RNA	volume for minimum RNA	Water up to 12 uL
2	1	250217EB	BP1h + 24R-1	107.7	1.75	2.08	1077	951	8.8	1.2
3	2		hypo1h + 24R-1	107.5	1.74	2.28	1075	951	8.8	1.2
4	3		hypo24h + 24R-1	95.1	1.77	0.13	951	951	10.0	0.0
5	4		BP1h + 24R-2	115.4	1.77	1.97	1154	951	8.2	1.8
6	5		hypo1h + 24R-2	129.1	1.76	2.13	1291	951	7.4	2.6
7	6		hypo24h + 24R-2	105.3	1.75	2.06	1053	951	9.0	1.0
8	7		BP1h	101.8	1.8	1.59	1018	951	9.3	0.7
9	8		hypo1h	112.6	1.74	2.19	1126	951	8.4	1.6
10	9		hypo24h	98.4	1.75	1.93	984	951	9.7	0.3

Dilute the cDNA 1:5 with nuclease-free water, to have 120 ul/sample total. -> store inn -20C freezer

2025年3月24日 星期一

qPCR

use 3 ul of DNA/qPCR reaction.
Prepare the table to guide the loading of the qPCR.
Use this as template:

250217 EB

- wt-11 BP+24R-1
- wt-11 1h Hypo +24R-1
- wt-11 24h Hypo +24R-1
- wt-11 BP+24R-2
- wt-11 1h Hypo +24R-2
- wt-11 24h Hypo +24R-2
- wt-11 BP
- wt-11 1h Hypo
- wt-11 24h Hypo

240603EB

- 4517 hypo 1h
- 5319 2.5mM
- 5319 hypo 1h
- 5319 hypo 24h

genes: STM2, STM2-cry, UNC13A, UNC13a-cry, ACTB, GAPDH

Entrer the number of samples in the yellow cell, number of genes in the green one, and prepare the qPCR mixes accordingly as follows:

QPCR materials:

1. Greiner Sapphire PCR consumable (384 Well PCR Microplates)
2. Film for 384 well plate (not aluminum or breathwasy film)
3. 8-channel (12-channel) auto-multichannel pipette (Boeke lab)
4. Reagents (RT-PCR box in -20A)
 - a. 2x master mix
 - b. nuclease-free water
5. Tips for multichannel (small, thin ones)
6. 25mL resevoir
7. primers (1:10) from the stock
(primer mixes are in the freezer in TC under bench)

1. setup tip spacing of 8-channel auto-multichannel pipette to 4.7 for 384 wells
2. load DNA with repeat dispense setting of 8-channel auto-multichannel pipette

- a. dispense: 3uL
 - b. first/last: 3uL
 - c. count: 14
 - d. save
 - e. run for every dispense
 - f. load until last one
 - g. purge run
3. add 17uL master mix for each gene with 8-channel (or maybe 12-well one) auto-multichannel pipette
 - a. setting: pipette -> edit -> adjust volume
4. Run the default on LC480
 - a. make sure the computer of **Felix&LC480** computer is on
 - b. press the corresponding bottom of the **Felix&LC480** computer (e.g., 4)
 - c. log into NYUMC account
 - d. Open LightCycler480 software
 - e. Login:
 - Username: admin
 - Password: LightCycler480
 - f. Put plate in machine by pressing the right button
 - g. Run from template like SYBR Green I
 - detection format: SYBR Green I/HRM Dye
 - check block size: 384
 - volume: 20
 - h. Save experiment
 - i. Start run
5. Export table from LightCycler480
 - a. click Analysis Tab
 - b. create new analysis: Abs Quant/Fit Points or Abs Quant/**2nd Derivative Max**
 - c. select all sample
 - d. click Calculate button at the bottom to calculate Cp number
 - e. Export the table as .txt

Analyze the experiment. You can use [this table](#) as a template.